ScamShield US: Agentic Retrieval-Augmented Generation for Consumer Fraud and Identity-Theft Education

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Introduction

This report presents *ScamShield US*, an open-source agentic retrieval-augmented generation (RAG) system for consumer fraud and identity-theft education. A user submits a suspicious message; the system classifies risk with Traditional, Advanced, and Hybrid natural language processing (NLP) models, retrieves the top five official guidance passages, and generates an educational answer with a local Qwen decoder (Qwen Team, 2024). Durable artifacts are stored in Amazon Simple Storage Service (S3) and accessed only through boto3; compute runs on Amazon Elastic Compute Cloud (EC2). The remainder of this report describes the data set, models, experimental setup, hyperparameters, results, conclusions, and references.

Description of the Data Set

Classification Corpora

We combine the UCI SMS Spam Collection (Almeida et al., n.d.) and the Super SMS Dataset (SMS Spam Research, n.d.). Files are stored under the course S3 prefix for raw data. After merge and deduplication, processed splits are written for training, validation, and testing. Binary ham/spam labels come from the sources; finer scam subtypes used in multi-class experiments are silver or heuristic labels where applied.

Knowledge Corpus for RAG

We seed 24 URL-backed official consumer-protection documents (for example, Federal Trade Commission, Cybersecurity and Infrastructure Security Agency, Federal Communications Commission, Social Security Administration, Internal Revenue Service, and Securities and Exchange Commission guidance) into the knowledge store with a provenance inventory. These documents are chunked and indexed for hybrid retrieval.

NLP Models and Algorithms

Traditional NLP

Messages are converted to term frequency–inverse document frequency (TF-IDF) features (with domain stopwords) plus lightweight linguistic and scam-signal features. We train logistic regression and a calibrated linear support vector machine, then majority-vote for the Traditional prediction. This corresponds to classical discriminative models from the course concept list.

Advanced NLP (Pretrained Transformer)

We fine-tune DistilBERT as a sequence classifier (Sanh et al., 2019). DistilBERT uses multi-head self-attention layers distilled from BERT, providing contextual token representations for scam-type discrimination.

Hybrid Cascade

Hybrid NLP keeps the Traditional label when the Traditional confidence margin is strong; otherwise it escalates to DistilBERT. The margin is selected on the validation split (final choice: 0.05).

Interpretability With LIME

To address course interpretability requirements, we apply LIME after binary spam/ham prediction (Ribeiro et al., 2016). The model predicts first; LIME then attributes which tokens support that prediction. Aggregated LIME weights are visualized as word clouds shown after the general corpus word clouds in the results dashboard and in this report.

Hybrid RAG and Decoder Language Model

Retrieval uses BM25 (sparse) and FAISS over MiniLM embeddings (dense; Reimers & Gurevych, 2019; Johnson et al., 2019), fused into a hybrid ranking with top_k = 5. The answer model is Qwen2.5-1.5B-Instruct, a decoder-only causal language model that predicts next tokens under a causal attention mask (Qwen Team, 2024). The prompt contains only retrieved official passages plus the user question; generation is greedy (do_sample = False).

Experimental Setup

Data are split into train, validation, and test sets with a fixed seed. Models are trained on the training split, the Hybrid margin is chosen on validation, and final comparison tables are computed on the test split. Implementation uses PyTorch and Hugging Face Transformers on an EC2 graphics processing unit, scikit-learn for classical models, and FastAPI with Angular for serving. Classifier performance is judged primarily by macro-averaged F1, with accuracy and weighted F1 as secondary metrics. Artifacts are uploaded to S3 via boto3.

Hyperparameters and Overfitting Controls

DistilBERT sequence length is capped (for example, 160 tokens during evaluation scoring), with AdamW optimization via Transformers Trainer defaults unless overridden in the training script. Hybrid margins searched on validation were {0.05, 0.10, 0.15, 0.20, 0.30}; 0.05 was selected. RAG chunk size, overlap, and hybrid fusion weight are configured in project settings, with final context limited to top_k = 5. Qwen uses a maximum of 320 new tokens with greedy decoding. Overfitting controls include never using the held-out test set for model selection, validation-based Hybrid margin selection, balanced class weights for classical models, and evaluation on the validation split during DistilBERT training where configured.

Results

Table 1 reports held-out test metrics. Hybrid NLP attains the best macro-averaged F1 (0.983) and accuracy (0.992), while routing only about 1.1% of test messages to DistilBERT. Traditional and Advanced NLP are both strong baselines; the cascade improves rare-class balance while keeping most traffic on the cheaper Traditional model. Dashboard visualizations include corpus word clouds (Figure 1) and LIME-based word clouds derived after model prediction (Figure 2).

Table 1

Held-Out Test Comparison of Classification Approaches

Approach	Macro-F1	Accuracy	Weighted-F1	Routed to advanced
Traditional NLP	0.969	0.987	0.987	0%
Advanced NLP	0.961	0.987	0.987	100%
Hybrid NLP	0.983	0.992	0.992	1.1%

Note. Metrics are computed on the held-out test split. Routing percentage is the share of messages escalated to DistilBERT under the Hybrid cascade.

Figure 1

General Corpus Word Clouds for Training Spam and Ham Messages



Note. Left panel shows spam term frequencies; right panel shows ham. These general word clouds reflect corpus document-frequency prominence before any model-explanation step. They are not DistilBERT attention maps.

After the corpus word clouds, the dashboard presents LIME (Local Interpretable Model-agnostic Explanations; Ribeiro et al., 2016) visualizations. For each sampled training message, a binary spam/ham TF-IDF + logistic regression model first produces a prediction; LIME then estimates which tokens pushed the predicted class probability up or down. Aggregating those post-prediction local weights yields LIME word clouds (Figure 2), which complement the frequency-based clouds in Figure 1.

Figure 2

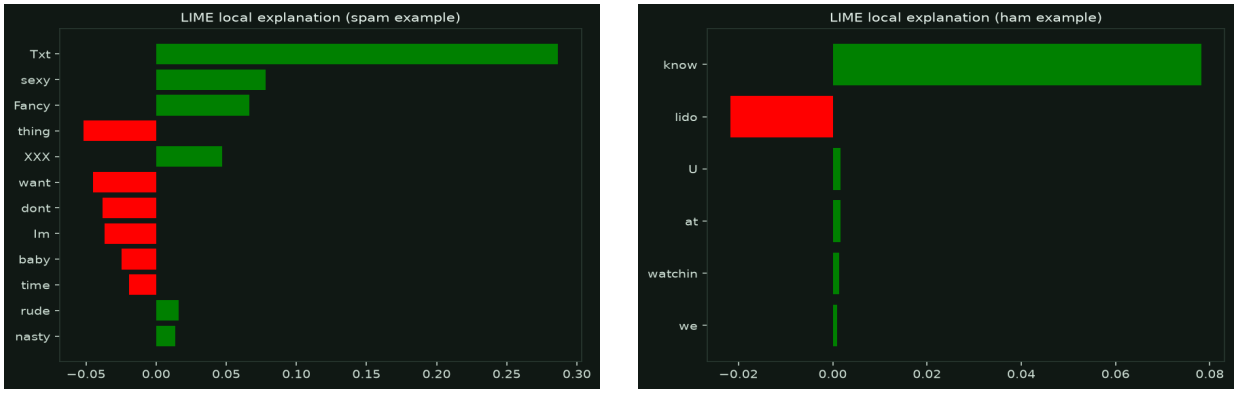
LIME Word Clouds After Model Prediction for Spam and Ham



Note. Left panel aggregates LIME feature weights for spam-class explanations; right panel aggregates ham-class explanations. Tokens are sized by mean absolute contribution after the classifier predicts the target class for each explained message.

Figure 3

Example Local LIME Explanations for Individual Spam and Ham Predictions



Note. Each bar chart shows token contributions for one predicted example (spam left, ham right). Positive weights support the explained class; negative weights oppose it.

The Angular user interface places the general word clouds first and the LIME word clouds immediately afterward, and also loads approach reports for confusion matrices and

per-label metrics from the same evaluation run.

Summary and Conclusions

We built an end-to-end educational fraud assistant combining classical NLP, DistilBERT, hybrid retrieval, a local Qwen decoder, and LIME interpretability under EC2 and S3 constraints. Hybrid cascading improved macro-averaged F1 while keeping most predictions on the cheaper Traditional model. Corpus word clouds summarize frequent terms, while LIME word clouds summarize tokens that matter after model prediction. Future improvements include SHAP or integrated gradients, optional recurrent network baselines, larger knowledge coverage, and higher-capacity open language models with quantization on A10G GPUs.

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